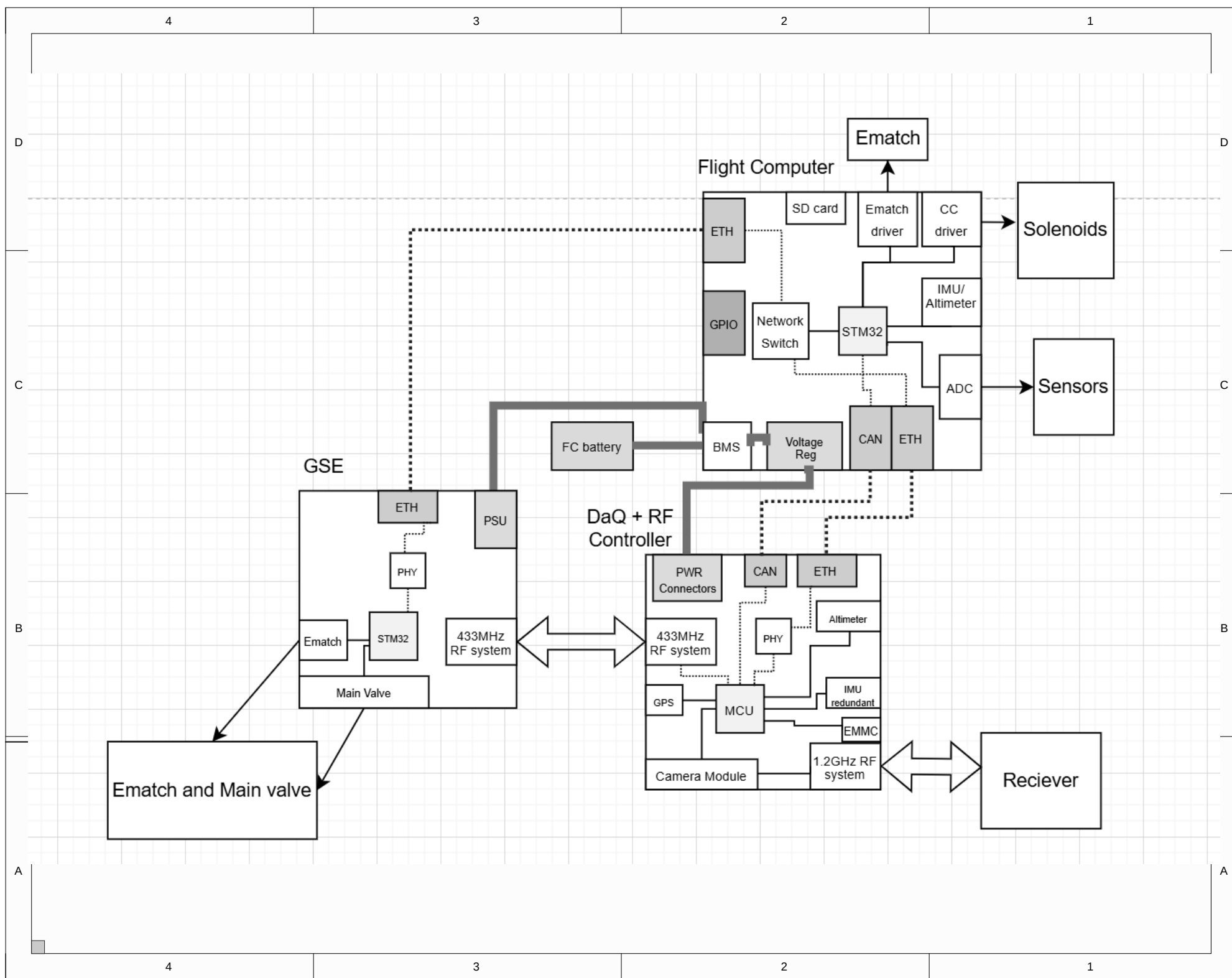
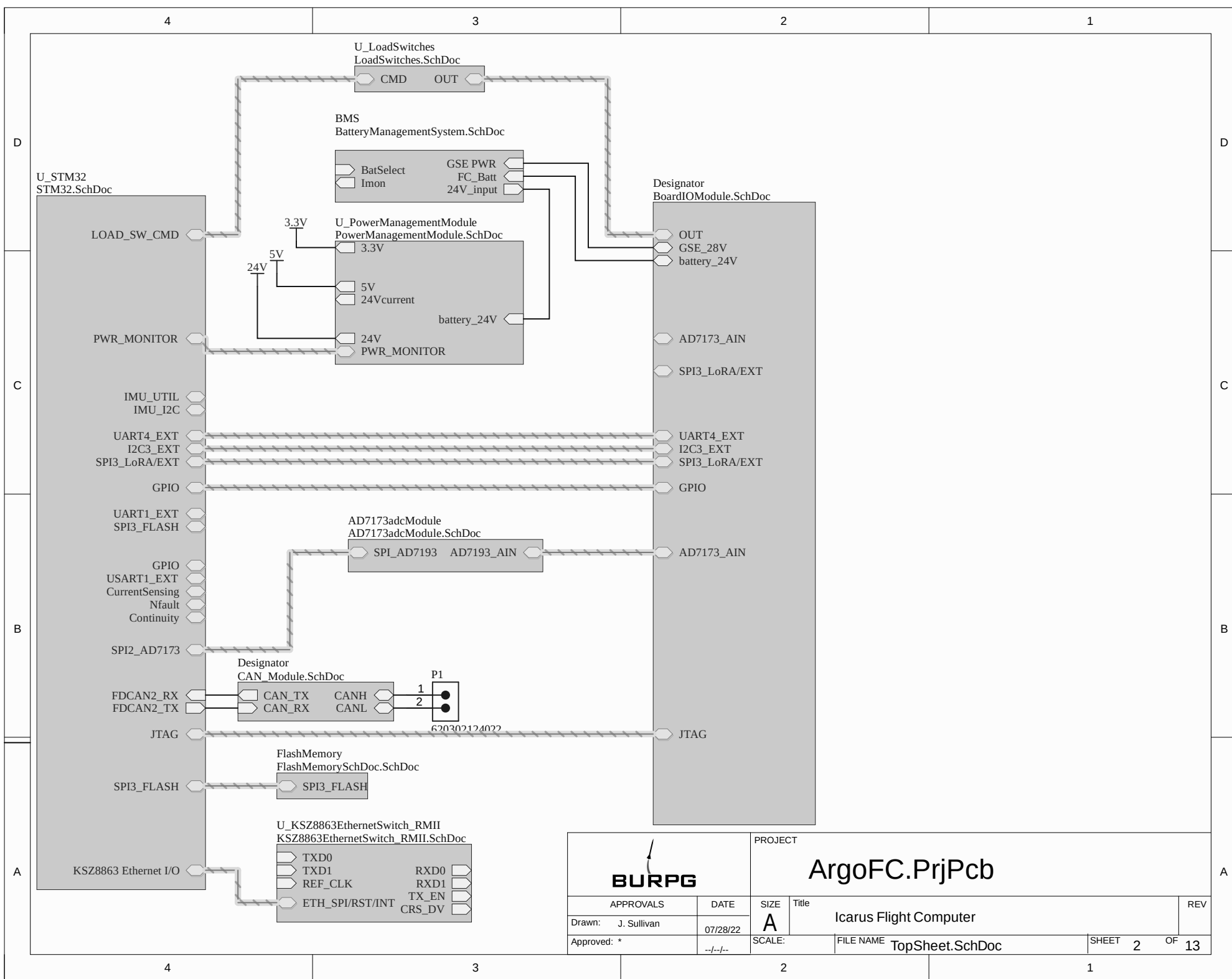






PROJECT

ArgoFC.PrjPcb

APPROVALS

DATE

SIZE

Title

REV

Drawn: J. Sullivan

07/28/22

A

Icarus Flight Computer

Approved: *

--/--

SCALE:

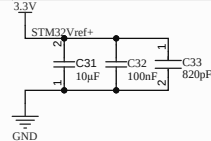
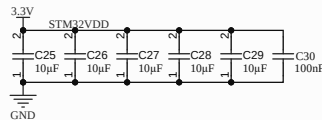
FILE NAME TopSheet.SchDoc

SHEET 2 OF 13

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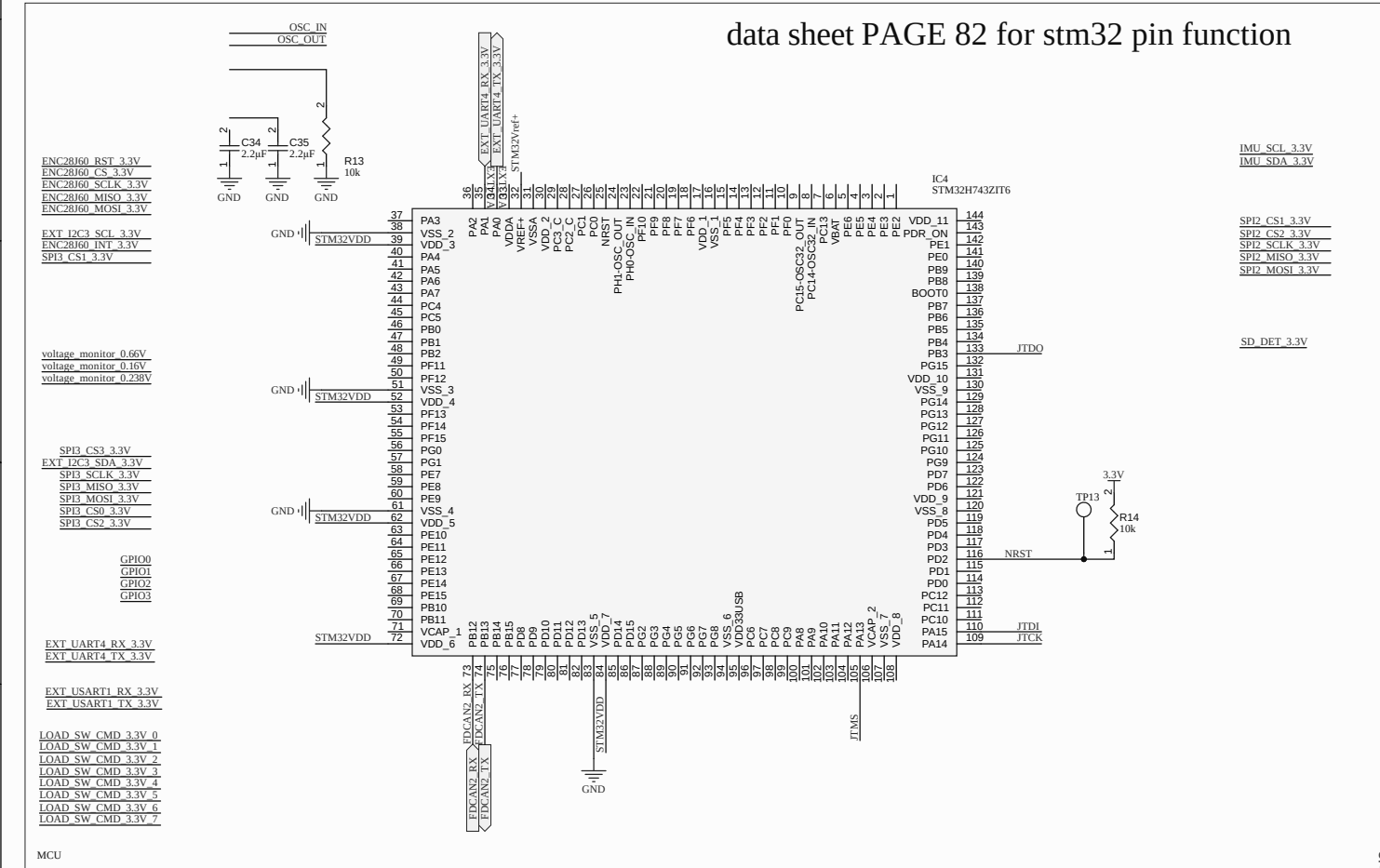


I will add a lot of cap here



I will add a lot of cap here

I deleted caps here for layout sanity, we should add them back later
data sheet PAGE 82 for stm32 pin function



ENC28J60 RST 3.3V
ENC28J60 CS 3.3V
ENC28J60 SCLK 3.3V
ENC28J60 MISO 3.3V
ENC28J60 MOSI 3.3V

EXT I2C3 SCL 3.3V
ENC28J60 INT 3.3V
SPI3 CS1 3.3V

voltage_monitor 0.66V
voltage_monitor 0.16V
voltage_monitor 0.238V

SPI3 CS3 3.3V
EXT I2C3 SDA 3.3V
SPI3 SCLK 3.3V
SPI3 MISO 3.3V
SPI3 MOSI 3.3V
SPI3 CS0 3.3V
SPI3 CS2 3.3V

GPIO0
GPIO1
GPIO2
GPIO3

EXT UART4 RX 3.3V
EXT UART4 TX 3.3V

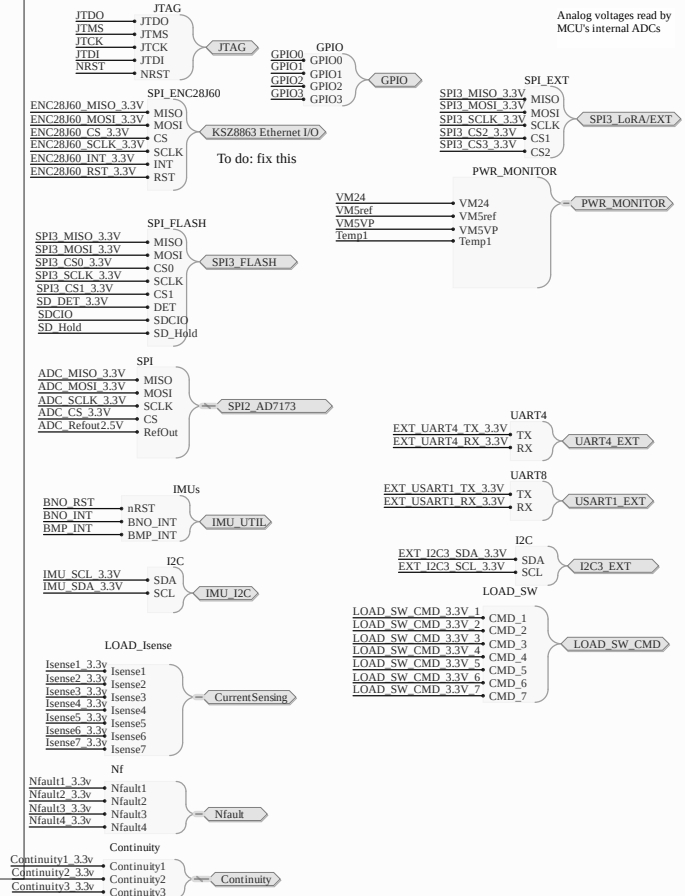
EXT USART1 RX 3.3V
EXT USART1 TX 3.3V

LOAD SW CMD 3.3V 0
LOAD SW CMD 3.3V 1
LOAD SW CMD 3.3V 2
LOAD SW CMD 3.3V 3
LOAD SW CMD 3.3V 4
LOAD SW CMD 3.3V 5
LOAD SW CMD 3.3V 6
LOAD SW CMD 3.3V 7

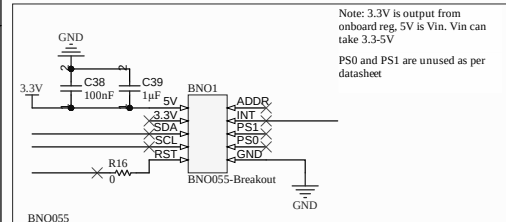
IMU_SCL 3.3V
IMU_SDA 3.3V

SPI2 CS1 3.3V
SPI2 CS2 3.3V
SPI2 SCLK 3.3V
SPI2 MISO 3.3V
SPI2 MOSI 3.3V

SD_DET 3.3V



To do: swap it to MS5607, might need to create a part for it



Memory mapping

Refer to Table 7 for details on STM32H742x/G flash and SRAM block memory mapping and to the product line reference manual for information on the boundary addresses for all STM32H742x/G peripherals.

APPROVALS		DATE	PROJECT
ENG:	*	12/01/2022	*
DSN:	J. Sullivan	-/-	PROJECT REVISION
CHK:	*	-/-	DOCUMENT REVISION
REFERENCE DOCUMENTS		TITLE	
BOM:		Icarus MCU	
ASSY DWG:	SIZE	CAGE CODE	DWG NO.
FAB DWG:	B		
PCB DWG:	SCALE	FILENAME	SHEET 3 OF 13

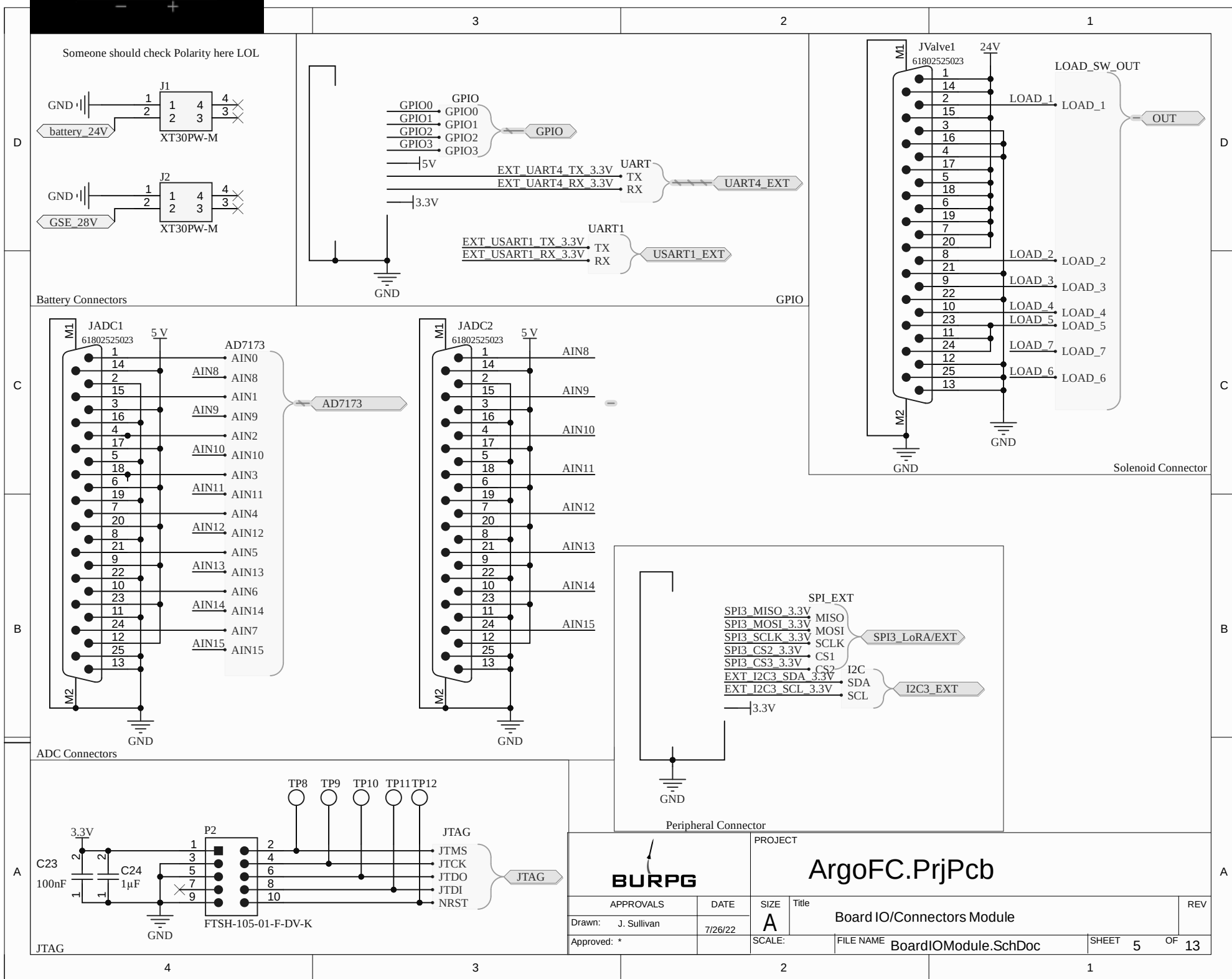


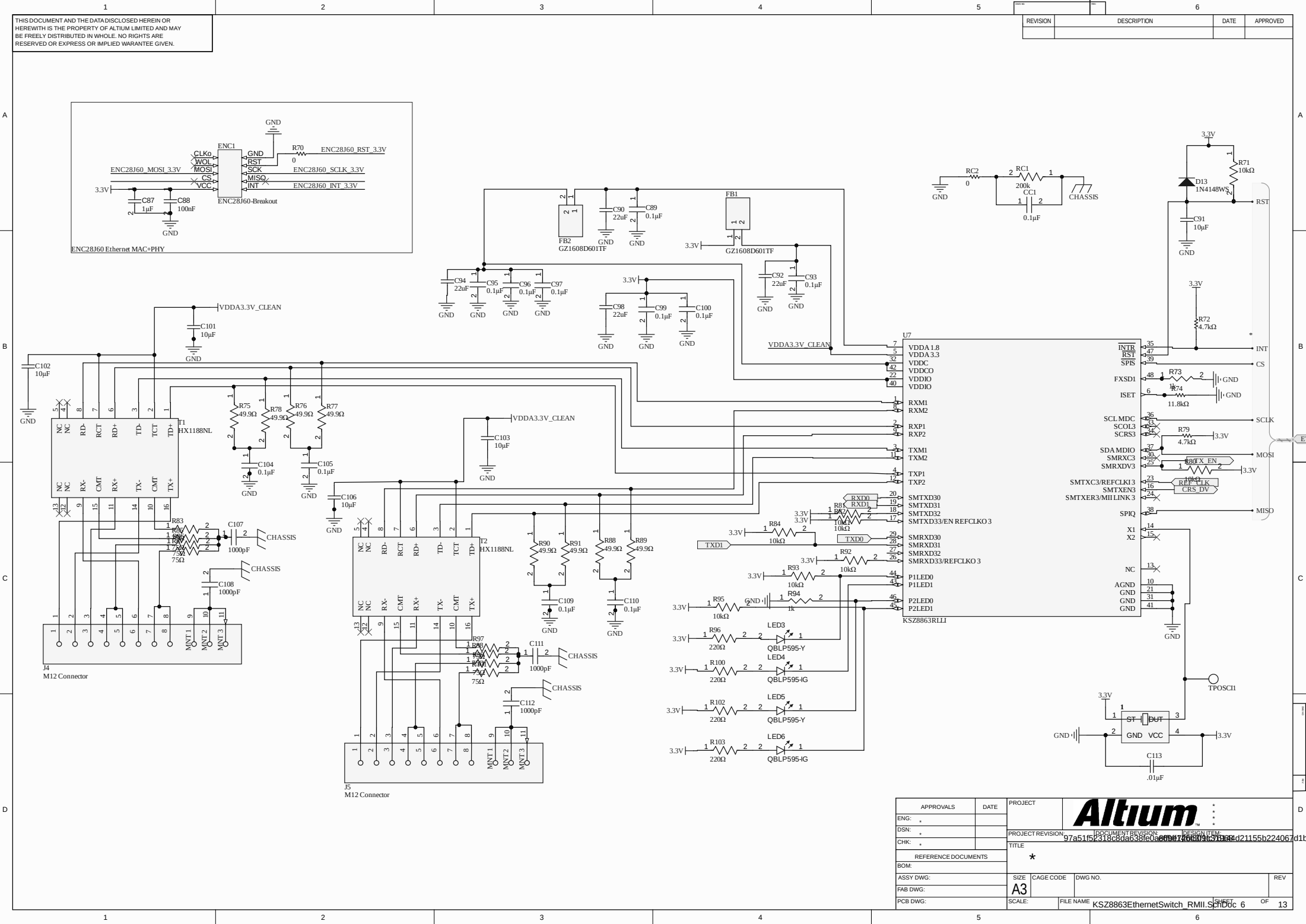
SPI

MISO 3.3V	→	MISO
MOSI 3.3V	→	MOSI
SCLK 3.3V	→	SCLK
CS 3.3V	→	CS
Refout 2.5V	→	

SPI2_AD7173

APPROVALS		DATE				
ENG:		+				
DSN:		+				
CHK:		+				
PROJECT REVISION			PROJECT REVISION: 1 97a51f52318c0da6538e0e4f4120c0c20155d21155b224067d1d1			
TITLE			* BOM:			
REFERENCE DOCUMENTS			ASSY DWG: FAB DWG: PCB DWG:			
BOM:			SIZE: CAGE CODE: DWG NO: REV:			
SCALE:			FILE NAME:		SHEET 4 OF 13	

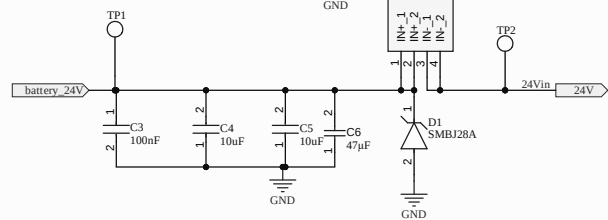




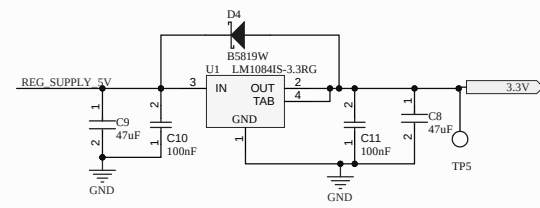
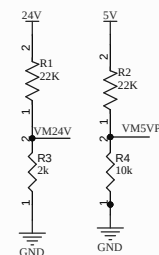
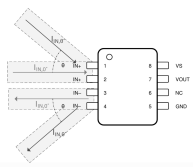
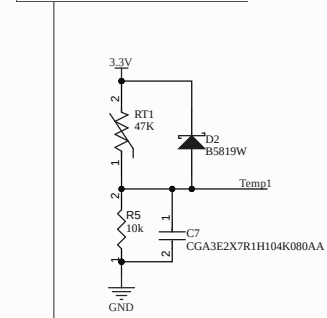
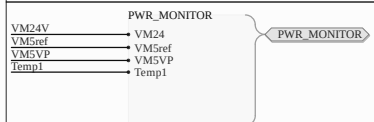
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REVISION	DESCRIPTION	DATE	APPROVED

APPROVALS		DATE	PROJECT		Altium	
ENG: *			PROJECT REVISION:			
DSN: *			DOCUMENT REVISION:			
CHK: *			DESIGN ITEM:			
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REFERENCE DOCUMENTS			TITLE			
BOM:			★			
ASSY DWG:			SIZE	CAGE CODE	DWG NO.	REV
FAB DWG:			A3			
PCB DWG:			SCALE:	FILE NAME	KSZ8863EthernetSwitch_RMII_SchDoc	SHEET 6 OF 13



Description
floating.
Connect a resistor Rset from m limit: $I_{lim} (A) = 6800 / Rset$ (oh



3.3V Regulator

over operating free-air temperature range (unless otherwise noted) ⁽¹⁾

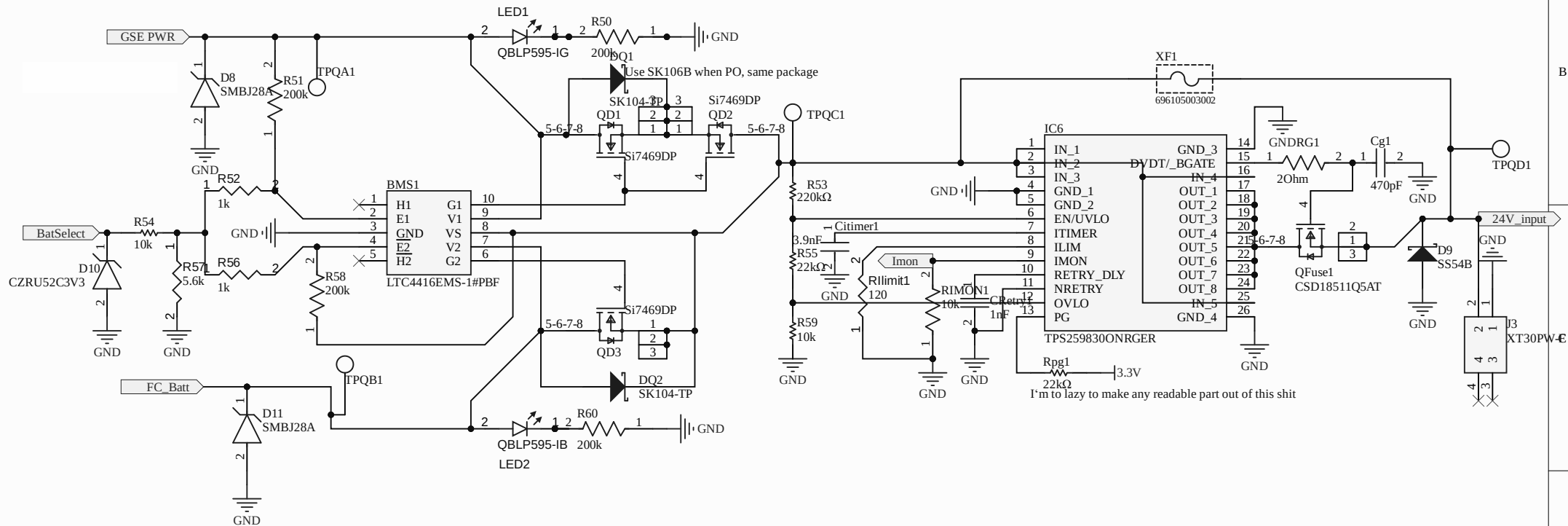
After operating free air temperature range (unless otherwise noted)		MIN	MAX	UNIT
Input voltage	VIN	-0.3	30	V
	EN	-0.3	6	
	BOOT		38	
	VSENSE	-0.3	3	
	COMP	-0.3	3	
	SS	-0.3	3	
Output voltage	BOOT-PH		8	V
	PH (10-ns transient from ground to negative peak)	-0.6	30	
Source current			-5	μA
	EN		100	
	BOOT		100	
	VSENSE		10	
Sink current	PH		9	A
	COMP		9	A
	SS		100	μA
Operating junction temperature, T _J		-40	150	°C
Storage temperature range, T _{stg}		-65	150	°C

Maximum Average Forward Rectified Current @60Hz sine wave, Resistance load TL (Fig.1)	I_{RAF}	A	10
Non-repetitive Peak Forward Surge Current @t=8.3ms Half-sine wave	I_{RSM}	A	175

Table 1. LTC4416 Operational Modes

E1	E2	Operation Mode	$I_{G(OFF)1}$	$I_{G(OFF)2}$
1	0	Load Sharing	Enabled	Enabled
1	Sense	V1 is Less Than V2	Enabled	
Sense	0	V1 is Greater Than V2		Enabled
0	X	Channel 1 Disabled. Do Not Use	Disabled	
X	1	Channel 2 Disabled. Do Not Use		Disabled
0	1	Both Channels Disabled	Disabled	Disabled

by default, V1 is enabled, V2 is disabled,
Pull-up resistor (E1) Pull-down resistor (E2)



For mostfet, I used SI7463DP for package and footprint, please use SI7469 when PO the board to save cost

Someone Should Double Check SOA for me



P-Channel 80-V (D-S) MOSFET

SI7469DP
Vishay Siliconix



www.vishay.com

P-Channel 40 V (D-S) MOSFET

SI7463DP
Vishay Siliconix

PRODUCT SUMMARY	Features
V _{DS} (V)	Power (W)
R _{DS(on)} (Ω)	I _D (A)

• TrenchFET™ Power MOSFETs



Title

Size

A4

Number

ABSOLUTE MAXIMUM RATINGS $T_A = 25^\circ\text{C}$

Parameter

Drain-Source Voltage

Gate-Source Voltage

Continuous Drain Current ($T_J = 150^\circ\text{C}$)

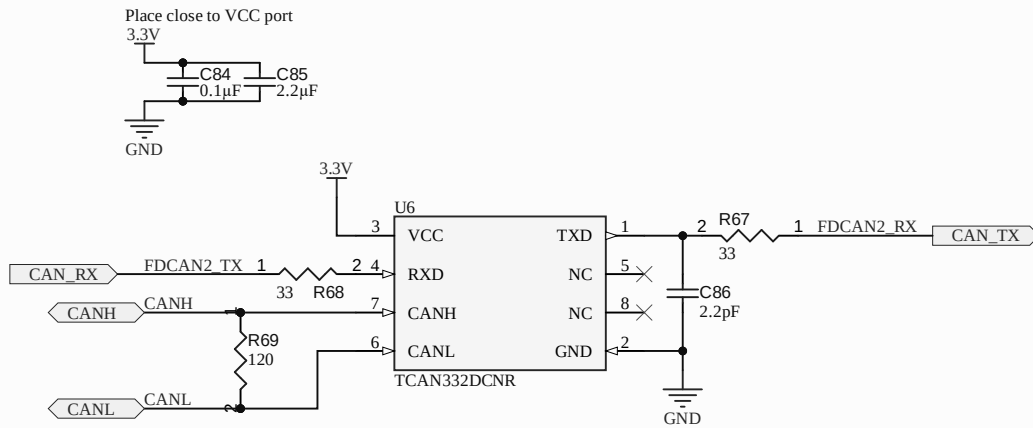
Pulsed Drain Current

$T_C = 25^\circ\text{C}$

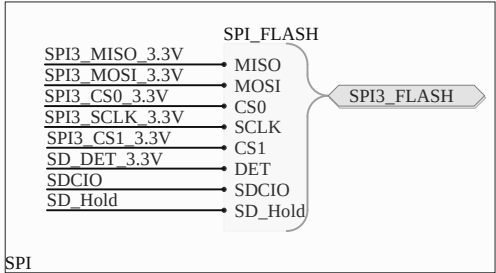
$T_C = 75^\circ\text{C}$

$T_A = 25^\circ\text{C}$

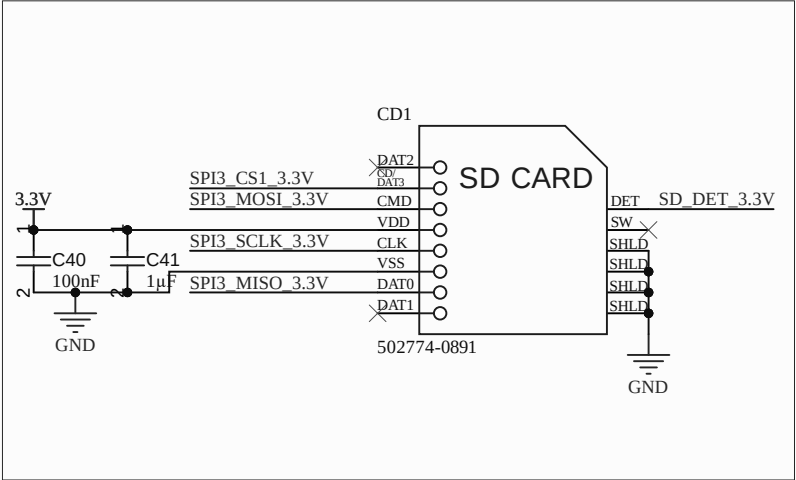
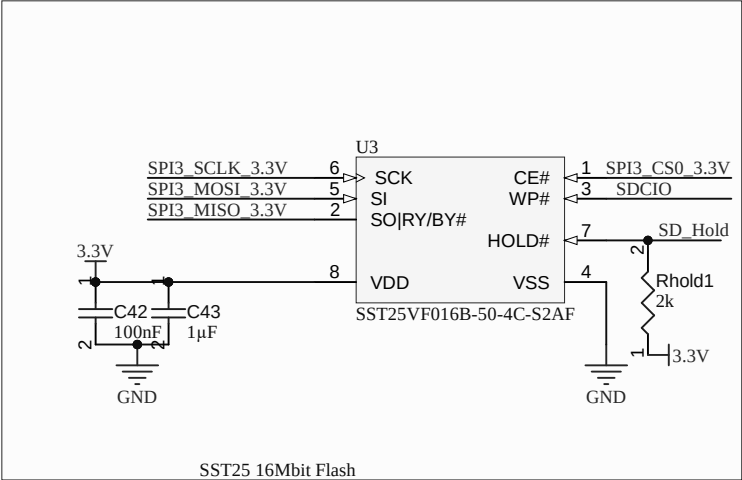
$T_A = 75^\circ\text{C}$



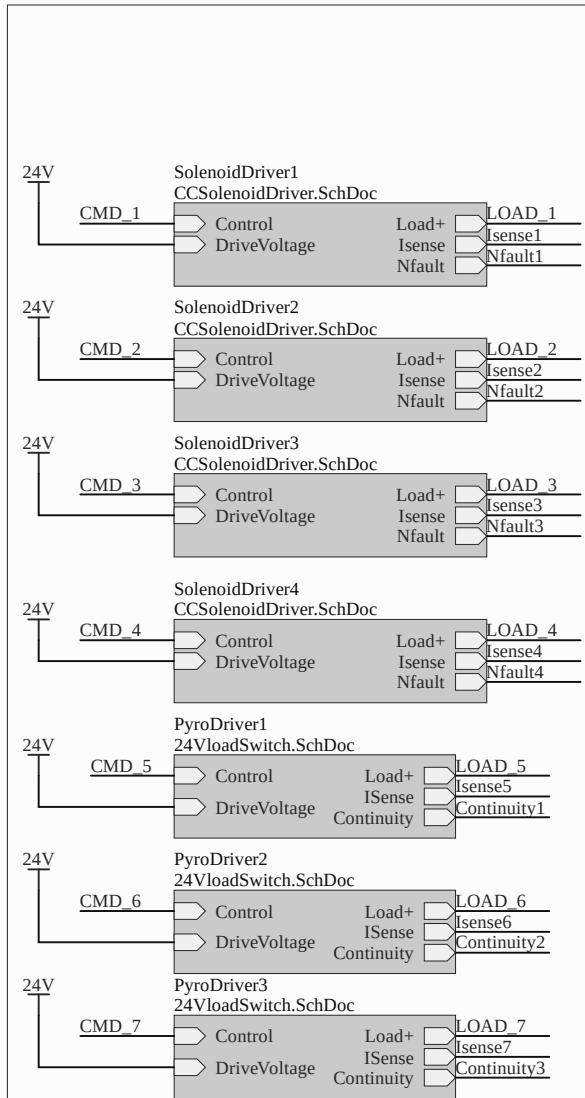
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Size A4	Number	Revision
Date:	8/03/2025	Sheet of
File:	C:\Users\...\CAN_Module.SchDoc	Drawn By:



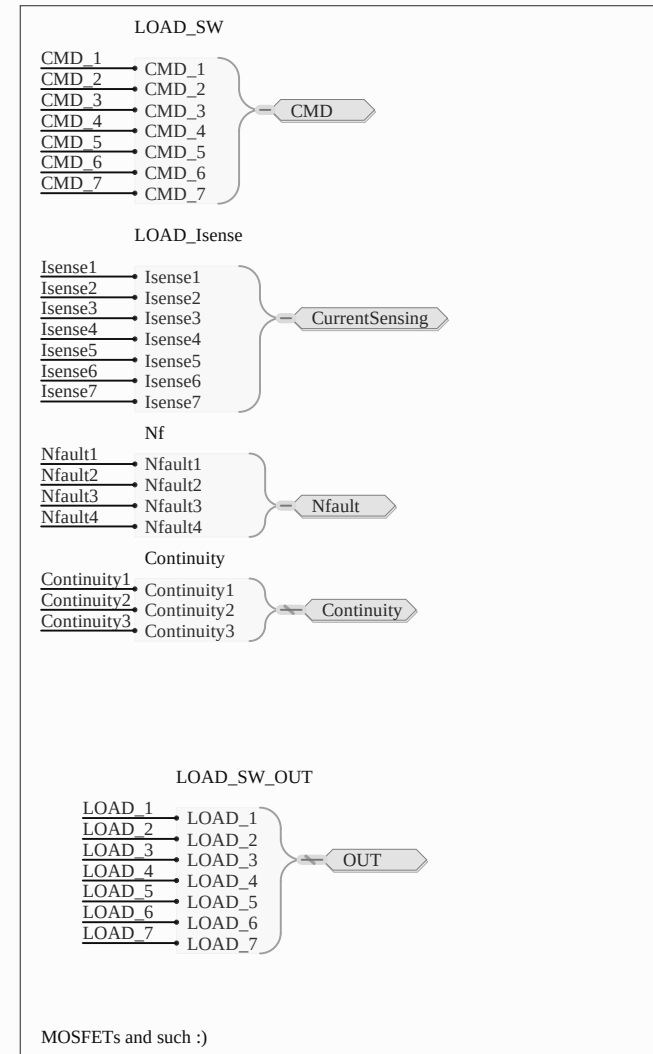
SPI



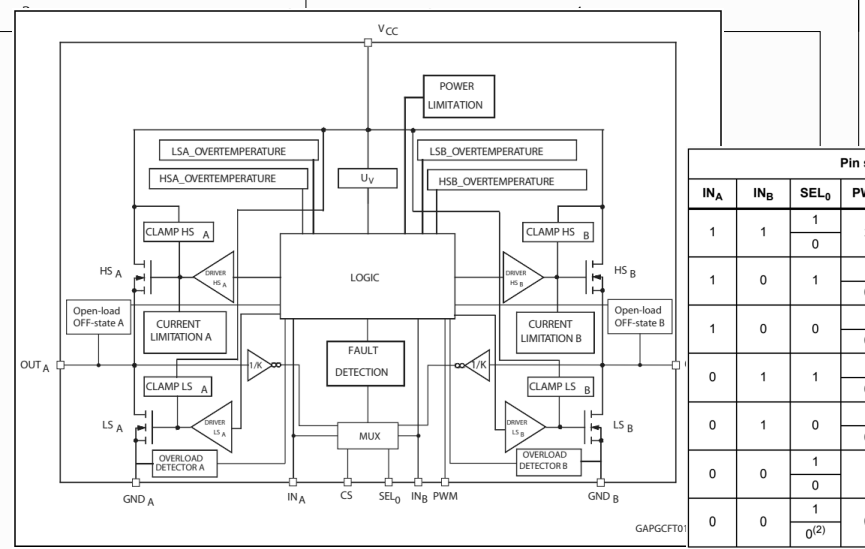
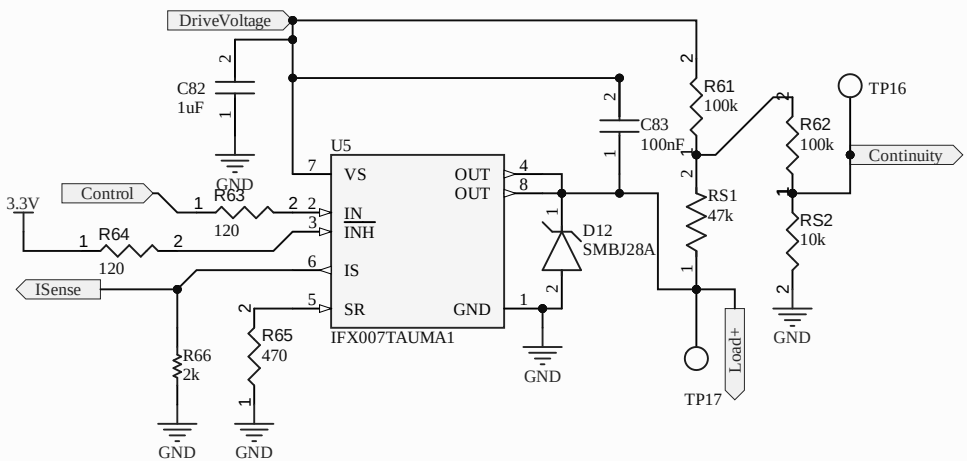
		PROJECT			
		ArgoFC.PrjPcb			
APPROVALS	DATE	SIZE	Title		REV
Drawn: *		A	*		
Approved: *		SCALE:	FILE NAME	FlashMemorySchDoc.SchDoc	SHEET 10 OF 13



MOSFETs and such :)



		PROJECT			ArgoFC.PrjPcb	
		APPROVALS	DATE	SIZE	Title	REV
		Drawn: *		A	Load Switches	
		Approved: *		SCALE:	FILE NAME LoadSwitches.SchDoc	SHEET 11 OF 13



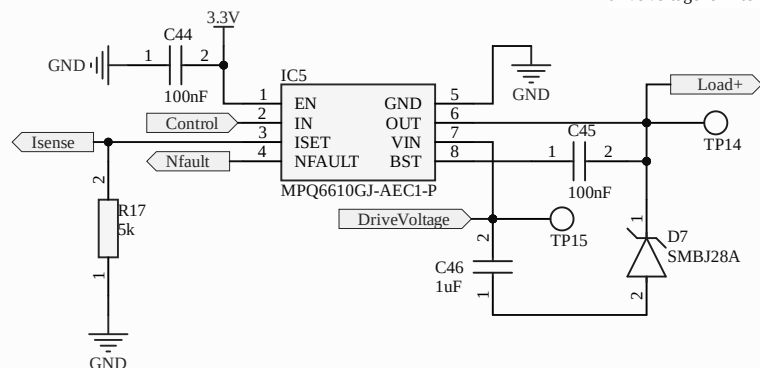
Pin st				
IN _A	IN _B	SEL ₀	PWM	
1	1	1	1	x
		0		
1	0	1	1	0
		0		
1	0	0	1	0
		0		
0	1	1	1	0
		0		
0	1	0	1	0
		0		
0	0	1	1	1
		0		
0	0	1	0	0
		0(2)		

Pin N°	Symbol	Function
1, 16	GND _A	Source of low-side switch A
2, 15	OUT _A	Source of high-side switch A / drain of low-side switch A
3	IN _A	Clockwise input
4, 5, 12	V _{CC}	Power supply voltage
6	IN _B	Counter clockwise input
7, 10	OUT _B	Source of high-side switch B / drain of low-side switch B
8, 9	GND _B	Source of low-side switch B
11	PWM	Voltage controlled input pin with hysteresis, CMOS compatible. Gates of low-side FETS get modulated by the PWM signal during their on phase allowing speed control of the motor
13	CS	Multiplexed analog sense output pin; it delivers a current proportional to the motor current according to the leg selection.
14	SEL ₀	Active high compatible with 3 V and 5 V CMOS outputs pin; in combination with IN _A , IN _B , it addresses the CurrentSense information delivered to the micro according to the operative truth table.

Title		
Size	Number	Revision
A4		
Date:	8/03/2025	Sheet of
File:	C:\Users\...\24VloadSwitch.SchDoc	Drawn By:

switch to MPQ6610GJ, same IC but TSOT-23-8 package and PCBA-able in JLCPCB

The IN pin is pulled down with an internal resistor.



C70: To generate the high-side gate drive voltage for internal Mosfet

The current limit triggered at ISET pin=1.5V

24V High Side Solenoid Driver

Input Logic

Each MOSFET in the MPQ6610 is controlled independently using the IN and EN pins (see Table 1).

Table 1: Input Logic Table

EN	IN	OUT
0	0	Hi-Z
0	1	Hi-Z
1	0	Low
1	1	High

The ISET pin voltage scaling is set by a resistor connected between the ISET pin and ground. For 1A of output current, 100μA of current is sourced into the resistor connected to ISET. For example, if a 10kΩ resistor is connected between ISET and ground, the output voltage on the ISET pin is 1V/A of output current.

The current is sensed anytime that either the LS-FET or HS-FET is on.

PIN FUNCTIONS

Pin # (TSOT23)	Pin # (SOIC-8)	Name	Description
1	8	EN	H-bridge enable input. Drive EN high to enable the half-bridge driver. The EN pin is pulled down with an internal resistor.
2	7	IN	H-bridge input pin. Drive IN high to drive the output high. The IN pin is pulled down with an internal resistor.
3	6	ISET	Current configuration resistor. Connect a resistor from ISET to ground to set the current limit and ISET pin voltage scaling.
4	5	nFAULT	Fault indication. The nFAULT pin is an open drain. It is logic low if a fault condition (e.g. over-current protection [OCP], over-temperature protection [OTP], or open load) occurs.
5	4	GND	System ground connection.
6	3	OUT	Output terminal.
7	2	VIN	Input supply voltage. Decouple VIN to GND with a minimum 100nF ceramic capacitor.
8	1	BST	Bootstrap pin. Connect a 100nF capacitor to the BST and OUT pins.

FUNCTIONAL BLOCK DIAGRAM

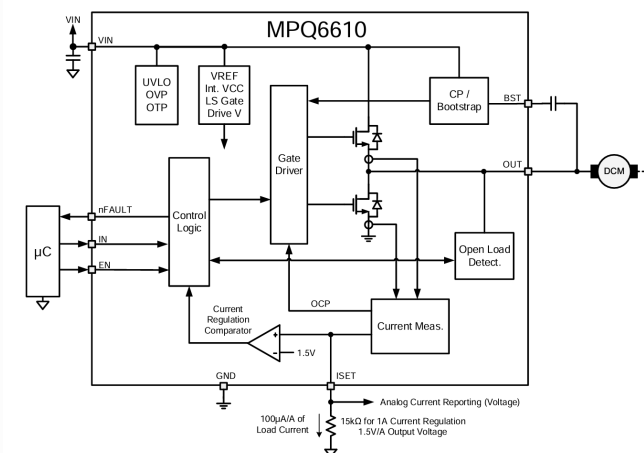


Figure 1: Functional Block Diagram

Title		
Size	Number	Revision
A4		
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File:	C:\Users\...\CCSolenoidDriver.SchDoc	Drawn By:

